

Miles Jennings

562-213-3013 | mjennin1@uci.edu | [linkedin.com/in/miles-jennings](https://www.linkedin.com/in/miles-jennings) | milesj-portfolio.app

EDUCATION

University of California Irvine

Irvine, CA

B.S Computer Engineering — Deans Honor List

Sept. 2022 – June 2026

Relevant Coursework: Organization of Digital Computers (Computer Architecture) Electronics I, Electronics II, Network Analysis I, Network Analysis II, Principles of Digital Design, Processor Hardware/Software Interfaces, Organization of Digital Computers

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

Test Engineer Intern

Buena Park, California

Leach International Corp.

March 2026 – Present

- Troubleshoot, made documentation for, and repaired test equipment by analyzing existing system architectures and adapting to legacy setups, eliminating production bottlenecks and ensuring on-time delivery
- **Developed** a **Python**-based capacitor test/conditioning system (LCR6002, High Voltage Power Supply, Safety Analyzer) for validation of **490** components, eliminating outsourced testing and reducing related company costs by **60%**
- Built and maintained hardware test setups using **oscilloscopes**, **multimeters**, and **power supplies** to perform functional and performance testing on electrical components

Control Systems Engineer

Irvine, California

HyperXite – SpaceX Hyperloop

Aug. 2025 – June 2026

- Designed, tested, and optimized STM32H753ZI-based control hardware in **C/C++**, interfacing with sensors via **UART**, **I2C**, and **CAN** while evaluating components for performance and cost improvements
- Built ESP32-based bidirectional communication system using ESP-NOW for real-time control and telemetry between pod and station and utilizing FreeRTOS to schedule critical tasks
- Designed a custom control board in **Altium** to integrate the pod's control hardware, collaborating with cross-functional teams to ensure reliable power distribution and acquisition of critical sensor and telemetry data

P2S Technology Intern

Long Beach, California

P2S Engineering

June 2025 – March 2026

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Improved design accuracy and project timelines by reducing regulatory revision cycles by **11%** through proactive 3D modeling and cross-functional coordination
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs to ensure reliable infrastructure design

PROJECTS

AudioVisor | C, nRF52840, PDM, I2C, Embedded Systems

Sept. 2025 – March 2026

- Led a team of 5 in the design and integration of an embedded audio-reactive wearable, coordinating hardware architecture, component selection, and firmware development
- Developed firmware in **C** on the **nRF52840** to interface with a digital microphone over **PDM**, performing real-time audio sampling and sound direction/amplitude detection
- Integrated peripheral communication and sensor interfaces using **I2C** and optimized embedded signal processing to achieve **84%** accuracy in sound localization during prototype testing

Embedded Building Management System | Python, Raspberry Pi, GPIO, LCD, DHT11, PIR

June 2025

- Developed an embedded control system on a Raspberry Pi integrating DHT11 and PIR sensors to autonomously manage HVAC, lighting, fire alarm, and security subsystems using **GPIO** interfaces
- Implemented **interrupt-driven** event handling, real-time sensor monitoring, and an LCD-based user interface with timestamped system logging for status reporting and diagnostics

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Verilog, RISC-V, MIPS

Software/Tools: LabVIEW, Revit, AutoCAD, Altium, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA